Student Performance Prediction System

AI / Machine Learning Internship Mini Project. Prepared by Mithraa RG

1. Introduction

This project builds a basic Machine Learning system that analyzes student-related data and predicts a student's final grade. It follows the complete ML workflow: data collection, cleaning, exploratory analysis, feature selection, training, testing, evaluation, and prediction.

2. Objective

To build and evaluate a model that predicts the final grade (G3, scored out of 20) from student background and study-habit data, and to create a simple prediction system that labels a student as Pass or At risk.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Google Colab (Jupyter Notebook). The project is hosted on GitHub at github.com/rgmithraa30/student-performance-prediction.

4. Dataset

The project uses the UCI Student Performance dataset (math course). It contains 395 students and 33 columns, including demographic details, family background, study time, past failures, absences, and three grades: G1, G2, and G3. Of the 33 columns, 16 are numeric and 17 are text (categorical). G3, the final grade, is the target variable.

5. Data Cleaning

The dataset was checked for missing values and duplicate rows. Both checks returned 0, so no values needed to be filled or removed. Data types were reviewed, and the text columns were marked for encoding before model training.

6. Exploratory Data Analysis

- **Grade distribution:** Final grades cluster around 10 to 12 in a bell shape, with a distinct spike at 0. These are likely students who dropped out or did not take the final exam.
- **Correlation with G3:** G2 (0.90) and G1 (0.80) are by far the strongest. Past failures show the strongest negative relationship (-0.36), followed by age (-0.16), going out (-0.13), and travel time (-0.12). Parents' education (Medu 0.22, Fedu 0.15) and study time (0.10) show weak positive relationships.
- **Study time vs. grade:** Students with higher study time had slightly higher median grades, but the effect was small.
- **Heatmap:** G1, G2, and G3 form a strong block of correlation. Mother's and father's education are related to each other, as are weekday and weekend alcohol consumption.

7. Feature Selection and Preparation

G1 and G2 are earlier-term grades that almost give away G3. To build a model that could work as an early-warning tool, they were removed from the main model. A second model that includes them

was trained for comparison. Text columns were converted to numbers using one-hot encoding, giving 39 input features for the main model.

8. Train-Test Split

The data was split 80% for training (316 students) and 20% for testing (79 students), with a fixed random state of 42 so results can be reproduced.

9. Model Training and Evaluation

Two regression models were trained: Linear Regression and Random Forest (100 trees). They were evaluated with R^2 (share of grade variation explained), MAE (average error in grade points), and RMSE (error that penalizes large mistakes more).

Model	R^2	MAE	RMSE
Linear Regression (without G1, G2)	0.141	3.395	4.196
Random Forest (without G1, G2)	0.245	3.148	3.935
Random Forest (with G1, G2)	0.815	1.165	1.949

Findings:

- Random Forest outperformed Linear Regression on every metric.
- Without earlier grades, the model explains about 25% of grade variation, with an average error of about 3 grade points. Grades are hard to predict from habits and background alone.
- Adding G1 and G2 raises R^2 to 0.815 and cuts the average error to about 1.2 points.
- **Feature importance:** In the model without G1 and G2, absences and past failures ranked highest, followed by health, going out, and age. The high rank of absences may be influenced by the students with a final grade of 0, so it should not be read as proof that absences directly cause lower grades.

10. Prediction System

A function, `predict_grade()`, takes a student's details (such as study time, failures, absences, and age), fills the remaining inputs with typical values from the training data, and returns a predicted final grade out of 20 using the Random Forest model. A grade of 10 or above is labeled Pass, and below 10 is labeled At risk. For example, a student with study time 3, no past failures, 2 absences, and age 17 was predicted to score 11.5 out of 20 (Pass).

11. Limitations

- The dataset is small (395 students) and covers one course at one point in time.
- The model without G1 and G2 has modest accuracy and should be used only as a rough early indicator.
- Students with a final grade of 0 are hard to predict and reduce model accuracy.
- Only default model settings were used. Hyperparameter tuning and cross-validation were not applied.

12. Conclusion

This project covered the full ML workflow on real student data. The results show that earlier grades are the strongest predictors of the final grade, while habit and background factors alone explain a smaller share. Random Forest gave the best results. Future work could include hyperparameter tuning, cross-validation, handling the zero-grade students separately, trying a Pass/Fail classification model, and building a simple web interface for the prediction system.